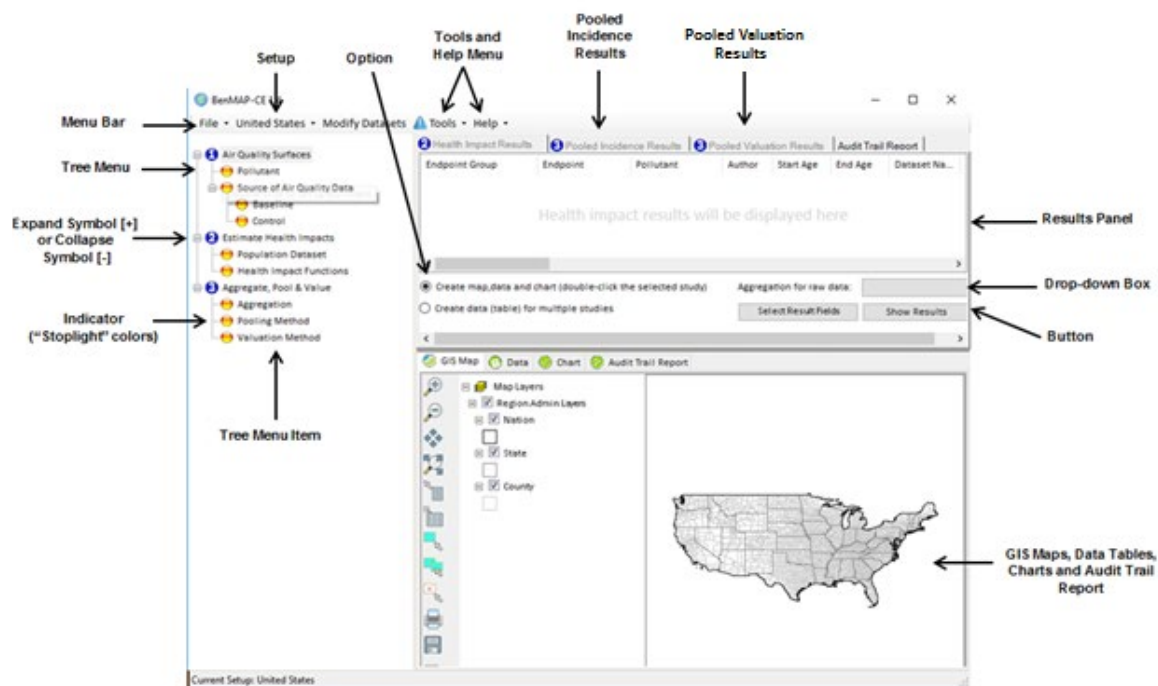
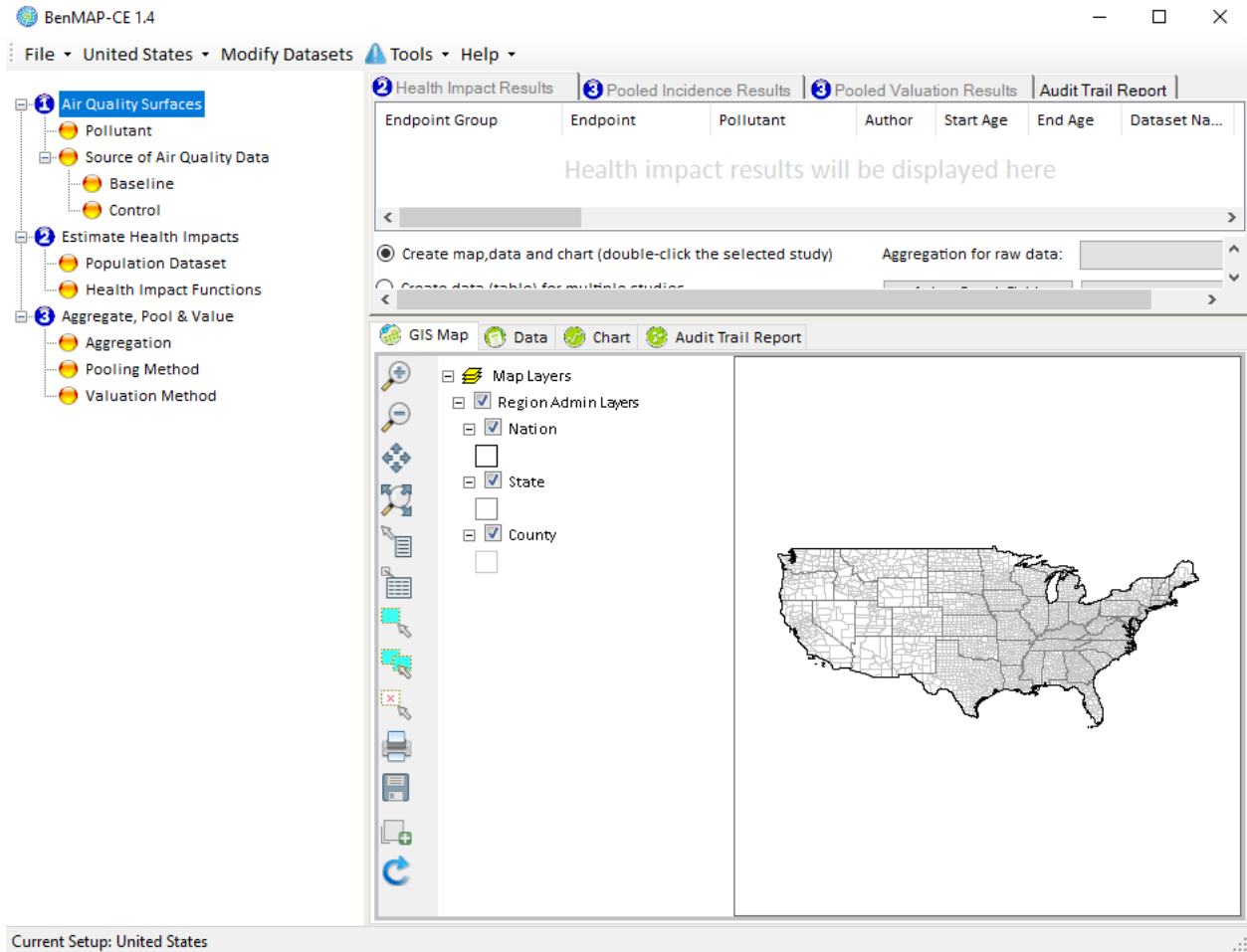




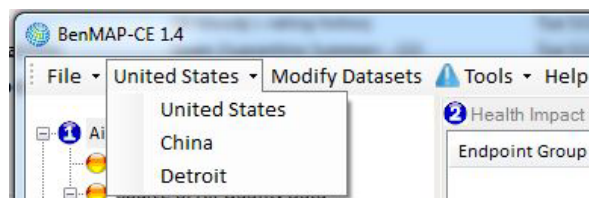


The tree menu on the left-hand window pane lists the analytical steps used in the Core Program. The tree menu items allow you to perform a highly customized health impact analysis. The **Tools** menu at the top of the screen is for less frequently used functions, such as importing and exporting data and special modules described in later sections.

The first section in this Chapter describes the Core Program features. The second section describes the additional functions found in the **Tools** and **Help** menus. The next section covers BenMAP-CE output options. Lastly, we answer some frequently asked questions. Note that this chapter provides an overview of functionality, not keystroke by keystroke instructions. Those detailed instructions may be found in Chapters 4 through 9.

3.1 Core Program

Beginning at the main BenMAP-CE window, you can choose the **Setup** you want to use for an analysis by selecting it from the **Setup** drop-down list. You are then ready to begin using the features available through the tree menu.



The tree menu takes you through the steps of an analysis. The first step, **Air Quality Surfaces**, allows you to select the pollutant of interest, and then specify the baseline and control air quality surfaces. The second step, **Estimate Health Impacts**, lets you choose the population dataset for a particular analysis, and then specify the health impact functions to estimate the incidence of adverse health effects. The last step, **Aggregate, Pool & Value**, gives you options for combining the health effects estimates and choosing economic valuation functions.

3.1.1 Create Air Quality Surfaces

BenMAP-CE is not an air quality model. Instead, it relies on externally-created air quality modeling data inputs. To estimate population exposure to air pollution, BenMAP-CE combines population data with air quality surfaces, which it generates using some combination of air quality modeling and/or monitoring data. In BenMAP-CE, air quality surfaces can be described as air quality grids (the structure) that have been populated with air pollution values (the data). The following is a brief description of each step. For detailed instructions, see Chapter 4: Loading Data and Chapter 5: Creating Air Quality Surfaces.

Pollutant

The **Pollutants** section of a setup specifies the pollutants that BenMAP-CE will analyze and defines the air quality metrics to be used by BenMAP-CE. You are not importing air pollution data, but rather naming your pollutants and defining the measures or metrics BenMAP-CE will use when assessing the health impacts associated with each pollutant.

Grid Definition

Air quality surfaces contain air pollution exposure estimates for a particular **Grid Definition**, as defined in the **Modify Datasets** window. **Grid Definitions** are typically comprised of either regularly shaped rectangles covering the region of analysis, or irregularly shaped polygons corresponding to political boundaries.

Modeling and Monitoring Data

To generate air quality grids, you can use air quality modeling data and air quality monitoring data in three different ways, as discussed below. However, once generated, all air quality grids have the same structure, and have the same *.aqgx extension that BenMAP-CE uses to designate these file types.

- **Model Data.** This procedure simply takes raw model data and converts it into a file that BenMAP-CE recognizes as an air quality grid. Defining the grid this way allows you to directly specify the air pollution values for each grid cell in a Grid Definition.
- **Monitor Data.** This procedure uses air pollution monitoring data to estimate air pollution levels for each grid cell in the selected Grid Definition. This may be done using one of three interpolation procedures - *Closest Monitor*, *Voronoi Neighborhood Averaging (VNA)*, or *Fixed Radius*. With closest monitor, BenMAP-CE simply uses the data of the monitor closest to each grid cell's centroid. With VNA, BenMAP-CE first identifies the set of monitors that most closely "surround" each grid cell, and then calculates an inverse-distance weighted average of the data from these neighboring monitors. With fixed radius, BenMAP-CE constrains the VNA interpolation to a user-specified distance around each monitor.
- **Monitor Rollback.** This procedure allows you to reduce, or roll back, monitor data using three methods: *Percentage Rollback*, *Incremental Rollback*, or *Rollback to a Standard*. Percentage rollback reduces all monitor observations by the same percentage. Incremental rollback reduces all observations by the same increment. Rollback to a standard reduces monitor observations so that they just meet a specified standard. After the monitor data is rolled back, it may be directly interpolated (as in *Monitor Data* grid creation) or combined with modeling data. This approach is described in more detail in Chapter 5: Creating Air Quality Grids, as well as in the Appendix A: Monitor Rollback Algorithms.

Once an air quality grid is created, it can be saved as a .aqgx file and reopened without any additional inputs needed.

